Aryaman Raj Jaiswal

+91 9136816530 | aryamanj250@gmail.com | linkedin.com/in/aryamanrajjaiswal
github.com/aryamanraj2 | aryamanj.in

Education

Netaji Subhas University of Technology — Dwarka, Delhi
Bachelor of Technology in Electronics and Communications — *Aug. 2024 – Present*

Experience

Bharti Airtel Limited — June 2026 – Aug. 2026
Software Engineer Intern — *Gurugram, India*

- Built an **agentic natural-language interface** over a production VM database: a **LangChain**-orchestrated **read-only text-to-SQL agent** behind **NeMo Guardrails** input/output rails, served over a **REST API**, running on self-hosted **Ollama** LLMs with **Vertex AI** for the heavier model path.
- Shipped an **AI surge-analysis** workflow that flags anomalous provisioning spikes across the fleet and summarizes probable root causes for on-call engineers.
- Re-architected the internal **OpenStack** platform tracking live and historical state across a distributed VM fleet; cut dashboard query latency by **30%** via **database migration**, **composite indexing**, and **cursor-based pagination**.

Examination Cell, NSUT — May 2025 – Aug. 2025
Software Developer Intern — *Delhi, India*

- Architected a **Next.js** front end over a **Flask REST API** and relational schema serving **10,000+ students**, cutting manual data entry by **70%** with **Pandas**-driven **constraint solvers** for invigilator assignment and room allocation.
- Shipped **NSUTrack**, native **Android and iOS** clients centralizing schedules, attendance, and notices for **1,500+ students**, on **Firebase Auth** and **FCM** push with an offline-first cache refreshed by **WorkManager** background sync.

SmartOut Media (Canada) — Jan. 2026 – Present
Software Engineer (Contract) — *Remote*

- Built the **SmartOut** app from the ground up (**50K+ installs**) — an offline-first geospatial product over a bundled **SQLite** dataset and a **GRDB** store with versioned schema migrations and **456** indexed regulatory polygons.
- Instrumented the product with **Firebase Analytics (GA4)** and **Crashlytics**, then triaged and resolved live issues **end to end** — from UI symptom through the network layer to the underlying query.

Projects

Wandr | *Node.js, Cloudflare Durable Objects, Supabase Realtime, WebSockets, Google Places/Routes API, Swift* — 2026

- Engineered a **seven-stage agentic coordinator** (extract → normalize → research → resolve → curate → validate → schedule) in which the model only *ranks* pre-vetted candidates and **deterministic validation is the sole authority** permitted to emit a plan — eliminating hallucinated output by construction.
- Built a **real-time multiplayer backend** on **Cloudflare Durable Objects** and **Supabase Realtime**: join-code/QR sessions, **WebSocket** presence and broadcast tallies at **sub-second consistency**, and **quorum locking** under host authority.
- Implemented **hybrid retrieval** over live **Google Places + Routes REST APIs** behind a swappable `VenueResearching` interface, plus a **constraint-ladder resolver** that relaxes the least critical rule first while **never relaxing dietary or accessibility** limits, so planning cannot dead-end.

AquaSense | *Python, Flask REST API, LangChain, Gemini, Random Forest, Computer Vision, SwiftUI + Kotlin clients* — Jan. 2026

- Built a **multi-agent LangChain** layer routing between three specialists: a **health agent** over live telemetry, a **disease agent** running **computer-vision** diagnosis on fish images, and a **marketplace agent** turning any flagged condition into a curated in-app cart.
- Grounded every **Gemini** claim in **Random Forest** classifier output to eliminate hallucinated water-toxicity verdicts, served from a stateless **Flask REST API** to native **SwiftUI** and **Kotlin** clients.

Achievements

- **Apple Swift Student Challenge Winner (2026):** Selected among the top global student developers for **Saldo**, a personal finance app running receipt parsing and spend intelligence entirely offline on an **on-device LLM**, **Core ML**, and **Vision**.
- **Winner, MLH Brainwave 2.0 (DTU):** Led Team Aquacult to **1st place over 200+ teams**, sweeping **Best Overall Project**, **Best Use of Gemini**, and **Best UI/UX Runner-Up**.
- **2x Winner, HackonHills (2024, 2025):** Back-to-back **1st-place** finishes among **125+ teams**, building **Aqua-Sense** (AI/IoT) and **HimYatra** (Mobile/Web) end to end.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, SQL (PostgreSQL), Go, Kotlin, Swift, C/C++, HTML/CSS
Full Stack: React, Next.js, Node.js, Express, REST API design, WebSockets, Flask, FastAPI, PostgreSQL, SQLite, Firestore
AI / Agentic: LangChain, LangGraph, agentic workflows, multi-agent orchestration, RAG, Text-to-SQL, NeMo Guardrails, Ollama, Vertex AI, Gemini, Foundation Models (on-device)
Cloud & Tools: Google Cloud (Vertex AI, Firebase Auth/Firestore/FCM, Analytics), Docker, Supabase, Cloudflare Durable Objects, Git/GitHub, Postman, Xcode
CS Fundamentals: Data Structures & Algorithms, System Design, Distributed Systems, Database Design, OOP